



Debugging Lecture II

ECEN 3314, Fall 2026

John Hu

September 27, 2026





Common Mistakes in this Lab

Let's go through our RCT. Some are particularly relevant for this week's lab.

Symptom	Root Cause	Bug Ticket	Circuit	Contributor
Voltage amplitude on the oscilloscope is off	The probe (1x/10x) setting was different from the oscilloscope's setting.	2025-02-04	Any	Anonymous
Vout incorrect on the bench OR "floating point computation failed during matrix solution" in SPICE simulations	Positive and negative supplies for the chip are swapped OR pin-1 of chip rotated	2025-02-11	Inverting amplifier using opamp	Anonymous
SPICE netlisting error	A voltage source is named the same as another signal's name	2025-02-11	Any	Anonymous
Signal amplitude on oscilloscope differs from signal source's setting	Did not select HiZ or 50 Ohm correctly based on the load	2025-03-04	Any, diode-based voltage doubler	Anonymous
Output with no signal but all noise	Connectivity error: source resistor (Rs) not connected to signal generator but grounded	2025-03-04a	Diode-based voltage doubler	Anonymous
Voltage amplitude on oscilloscope off by 1000x	Used AC coupling instead of DC coupling	2025-03-04c	Diode-based voltage doubler	Gunnar



Common Circuit Mistake #3: Power and Ground

It may sound stupid, but **power** and **ground** should be among the first to check:

- Did you connect **all** the power? (Circuits may have multiple supply rails, such as V_{CC+} , V_{CC-} , $COMM$)
- Have you shorted **all** grounds (digital ground $DGND$, analog ground $AGND$, power ground $PGND$, equipment ground, and PCB/breadboard ground) together?
 - It is very common for commercial chips to have multiple grounds not internally connected
 - So you need to connect all GND together on your circuit

For more common mistakes, go to [this TI webpage](#) or our RCT on the forum.



Global Debugging Strategies

What are the general steps for circuit debugging?





Strategies of Bug Location

In analog circuit debugging, the most time consuming part is often **bug localization** (Mitra et al. 2010), or finding the “*smoking gun*” via a light switch test (Josephson 2002), i.e., turning the the failure “on and off” by **varying a condition**!

- Once the root cause is found, the repair is often straight forward

How to search for the root cause is the hard part! Some engineers refer it as **finding the needles in the haystack** (Josephson 2002).

- Experienced engineers often debug by experience, i.e., recalling previous bug episodes and applying similar solutions (Konradt 1995)
- Beginners, however, are still better off debugging **systematically** rather than trial-and-error (Böttcher et al. 2016)



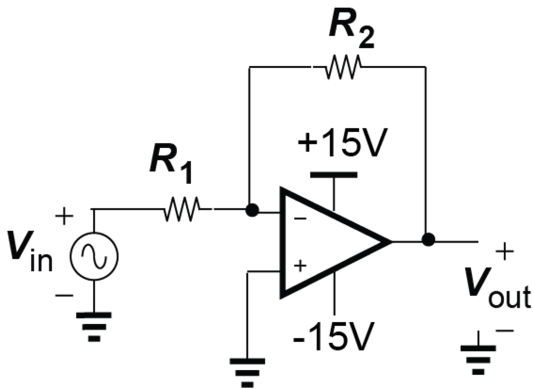
Hypothesis Generation

Hypothesis generation is central to bug localization. It requires “causal reasoning”:

- If H_i is the cause, will H_i cause the symptom(s) seen?
- If no, then H_i is likely not the cause.

Think of the next hypothesis, H_{i+1}

Let's **practice**. Assume in the following inverting amplifier, V_{out} becomes an in-phase sine wave with V_{in} (instead of out of phase). What could be a plausible root cause?

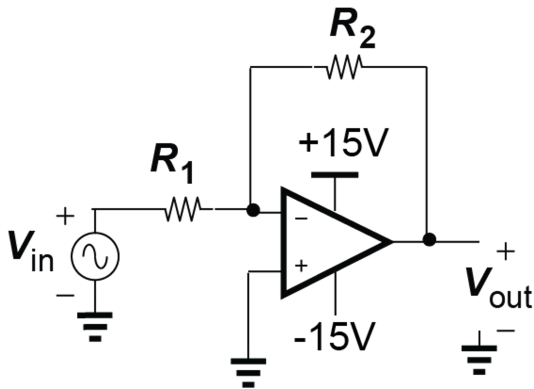




Hypothesis Generation

Plausible root causes:

1. The R_1 and R_2 connection was unexpected disconnected from V_{in+}

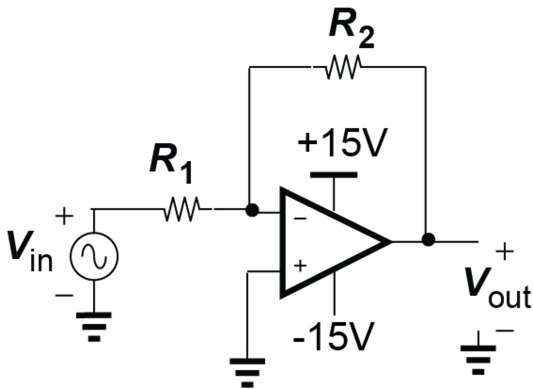




Hypothesis Generation

Plausible root causes:

1. The R_1 and R_2 connection was unexpected disconnected from V_{in+}
2. The output of the opamp is disconnected from V_{out}





Knowledge and Skills Required

The ability to generate high-quality hypotheses requires:

- Domain knowledge (This course: ECEN 3314)
- Functional understanding: functional model(s) of the circuit

Once H_i is generated, you need to test if H_i is indeed the true root cause. This will require:

- Testing knowledge: how to design an experiment to test H_i ? What are observable? What are controllable? What are repeatable?
- Procedural knowledge: how to use equipment?
- Hands-on skills

Debugging is an indicator of your **comprehensive** knowledge and skills.



References I

- Böttcher, Axel, Veronika Thurner, Kathrin Schlierkamp, and Daniela Zehetmeier (Oct. 2016). "Debugging students' debugging process". In: *2016 IEEE Frontiers in Educ. Conf. (FIE)*, pp. 1–7. DOI: 10.1109/FIE.2016.7757447.
- Josephson, D.D. (Oct. 2002). "The manic depression of microprocessor debug". In: *Proc. Int. Test Conf.* Pp. 657–663. DOI: 10.1109/TEST.2002.1041817.
- Konradt, Udo (Oct. 1995). "Strategies of failure diagnosis in computer-controlled manufacturing systems: empirical analysis and implications for the design of adaptive decision support systems". en. In: *Int. J. Human-Computer Studies* 43.4, pp. 503–521.
- Mitra, Subhasish, Sanjit A. Seshia, and Nicola Nicolici (June 2010). "Post-silicon validation opportunities, challenges and recent advances". In: *Proc. 47th Design Automat. Conf. (DAC)*, pp. 12–17.



Thank you!

John Hu
john.hu@okstate.edu